Predicting Mental Health Risk from Social Media Usage: A Multi-Dataset Pipeline with Cross-Dataset Evaluation

ELEC3544 Data Science with Foundation Models — Group O

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May 2026

Abstract

We present a pipeline for predicting mental health risk from social media usage behavior, integrating eight independently collected survey datasets with heterogeneous mental health instruments. The central challenge is constructing a unified target variable across instruments ranging from clinically validated scales (GAD-7, PHQ-9) to single-item self-reports. We address this through theoretical range normalization, which maps each variable to [0,1] using its instrument's design endpoints, preserving absolute severity semantics across datasets. A Random Forest classifier achieves F1 macro of 0.714 under standard cross-validation, with late-night usage, the screen-time-by-late-night interaction, and social comparison emerging as the most robust predictors across all eight datasets. However, Leave-One-Dataset-Out (LODO) validation reveals a sharp generalization gap (F1 macro = 0.272), traced to self-selection bias, where voluntary survey participants skew toward higher severity, producing fundamentally different target distributions between real and synthetic datasets. Reweighting real data recovers High Risk detection from F1 = 0.04 to 0.66. Throughout this work, a foundation model served as an implementation and analysis partner, with human judgment driving every design decision.

1. Introduction

The relationship between social media usage and mental health has become a significant public health concern, with growing evidence linking excessive use to anxiety, depression, and sleep disturbance [6]. A key practical question is whether observable usage behaviors, such as screen time, late-night browsing, and social comparison, can serve as predictive signals for mental health risk. If so, behavioral data that is readily available could inform early intervention without requiring clinical assessment.

A fundamental challenge in studying this question at scale is that mental health is measured differently across surveys. Clinically validated instruments like the GAD-7 [1] (anxiety, 0–21) and PHQ-9 [2] (depression, 0–27) coexist with custom Likert scales, happiness indices, and categorical self-reports. Integrating data from multiple such instruments requires a principled approach to target variable construction that preserves the severity semantics embedded in each instrument's design.

This project makes three contributions. First, we develop a target unification methodology based on theoretical range normalization and a feature-target separation principle that distinguishes subjective psychological states from observable behaviors. Second, through Leave-One-Dataset-Out (LODO) evaluation, we identify self-selection bias as the dominant source of distribution shift between real and synthetic survey data, a finding with implications for any multi-source mental health study. Third, we demonstrate that reweighting strategies can recover minority-class detection even when training data distributions differ substantially from test distributions.

The remainder of this report is organized as follows. Section 2 describes the eight datasets and data quality findings. Section 3 details our methodology, including target construction, imputation, and modeling. Section 4 presents results and analysis, with particular depth on cross-dataset generalization and distribution shift. Section 5 discusses findings, limitations, and implications.

2. Data Description

2.1 Dataset Overview

We collected eight datasets from Kaggle covering social media usage and mental health, comprising three real surveys and five synthetic datasets (Table 1). Together they contain 11,858 usable records. The datasets vary substantially in size (20 to 8,000 rows), population (university students, general adults, digital diet participants), and mental health instrumentation.

Table 1: Dataset summary. MH = mental health instrument(s) used.

#	Type	Rows	MH Variables	Instrument(s)
1	Real	49	1	Categorical self-report
2	Synth	2,000	4†	Custom 0–20, 1–10 scales
3	Synth	500	3	Custom 1–10 scales
4	Real	481	6	Likert 1–5 items
5	Synth	20	3	MHI, FOMO, productivity
6	Synth	8,000	2	GAD-7 (0–21), PHQ-9 (0–27)
7	Real	705	1	Custom 4–9 scale
8	Synth	103	1	Categorical emotion

†Originally 5; `mental_health_score` excluded (see §2.2).

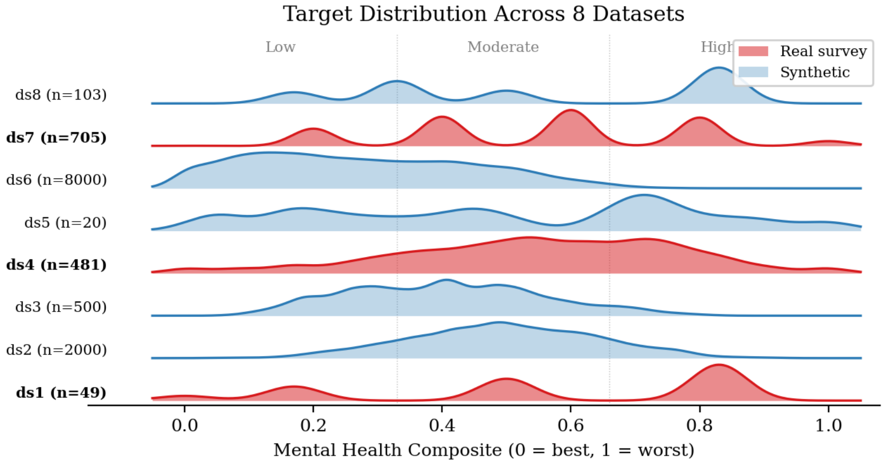

Figure 1: Target composite distributions across all eight datasets (ridgeline plot). Red shading denotes real surveys, blue denotes synthetic data. Dotted lines mark three-class cutoffs. Real surveys cluster at higher severity, while synthetic datasets peak near the Low–Moderate boundary.

2.2 Data Quality Findings

Dataset 2 anomaly. During systematic correlation analysis, we discovered that the `mental_health_score` variable in Dataset 2 (2,000 rows) exhibits near-zero Pearson correlation with every other variable in the dataset, including weekly anxiety ($r = 0.000$), depression and stress (both $r \approx 0.000$), and mood ($r = -0.028$). Its distribution is uniform across integer values 20–80, invariant across gender and location subgroups. This pattern is consistent with a randomly generated placeholder. We excluded it from the target composite while retaining the dataset's four valid mental health indicators. This check was performed systematically with the assistance of a foundation model, which audited pairwise correlations across all 20+ variables; the human team verified the distributional evidence and made the exclusion decision.

Structural missingness. Because datasets collected different feature sets, merging produces substantial structural missingness, with 36 of 48 features present in fewer than 20% of rows. Of the remaining features, age, gender, screen time, and platform are available in nearly all datasets, while others such as sleep duration (3/8), social comparison (2/8), and late-night usage (1/8) have partial coverage but carry strong predictive signal. This missingness pattern represents genuinely different survey designs, not data collection errors, and must be handled by imputation methods that respect inter-feature relationships rather than naive mean filling.

3. Methodology

3.1 Feature-Target Separation

We adopted a separation principle whereby variables measuring subjective psychological states (how one feels) are designated as target components, while variables measuring observable behaviors (what one does) serve as predictive features. This distinction reflects a practical inference gap, since in a deployment scenario, behavioral signals such as screen time or posting frequency can be observed directly, whereas internal states like anxiety or mood cannot. By predicting the latter from the former, the model addresses a genuine prediction task rather than a tautological one.

Borderline cases require explicit judgment. Sleep duration (an observable, measurable quantity) is classified as a feature, while sleep quality (a subjective self-rating) contributes to the target. Dataset 7's addiction score, despite its psychological component, is placed on the feature side because it primarily describes a behavioral usage pattern and because its extremely high correlation with the mental health score ($r = -0.945$) would dominate the composite if included as a target component. Stress level is classified as a target component because it represents an internal emotional state rather than an observable action. These decisions were made through iterative discussion, with the foundation model systematically enumerating and classifying all variables across eight datasets, and the human team defining the separation principle and adjudicating every borderline case.

3.2 Target Construction

Normalization. Each mental health variable is mapped to $[0, 1]$ using its instrument's theoretical (design) range, with direction unified so that higher values indicate worse mental health. Representative formulas are

$$\begin{aligned} \text{GAD-7:} \quad & x_{\text{norm}} = x/21 \\ \text{Happiness (1–10):} \quad & x_{\text{norm}} = (10 - x)/9 \\ \text{PHQ-9:} \quad & x_{\text{norm}} = x/27 \\ \text{Likert (1–5):} \quad & x_{\text{norm}} = (x - 1)/4 \end{aligned}$$

We selected theoretical range normalization over three alternatives. Observed min-max normalization is sample-dependent, so if a dataset contains only mild cases, it stretches mild scores to appear severe, undermining cross-dataset comparability. Quantile normalization forces a uniform distribution, destroying the clinical severity information embedded in the original distribution; if 66% of respondents genuinely report minimal anxiety, quantile normalization manufactures fine-grained distinctions where the instrument detected none. PCA-based composites fail for single-variable datasets and produce components whose meaning differs across datasets. This comparison was conducted systematically with the foundation model analyzing trade-offs for each approach; the human team evaluated the reasoning, pushed back on quantile normalization, arguing that distributional skew reflects genuine prevalence rather than measurement artifact, and made the final selection.

Composite and discretization. Normalized indicators within each dataset are averaged with equal weight, as no domain consensus supports differential weighting of mental health facets. The composite is discretized into three risk classes using fixed cutoffs of Low $[0, 0.33)$, Moderate $[0.33, 0.66)$, and High $[0.66, 1.0]$. Fixed cutoffs preserve the absolute severity semantics established by theoretical normalization. The resulting class distribution contains Low 5,626 (47%), Moderate 5,404 (46%), and High 828 (7%).

3.3 Data Processing Pipeline

Cleaning and alignment. Each dataset was cleaned independently, with column names standardized to lowercase snake_case, gender harmonized to {male, female, other}, platform names unified across datasets, and screen time converted to hours (categorical midpoints for Datasets 1 and 4, minutes-to-hours for Datasets 5 and 8). After per-dataset target construction, all datasets were merged via outer join, producing 11,858 rows and 48 columns. Features unique to specific datasets appear as structural NaN in others.

Imputation. Merging creates substantial structural missingness. We compared three imputation methods by their downstream impact on classification (Table 2). MICE (Multiple Imputation by Chained Equations) [4] iteratively predicts each missing feature from the others, capturing inter-feature relationships that simpler methods ignore. For context, a prior analysis of this data used mean imputation, which collapsed 78% of `social_media_hours` values to a single value of 2.334, effectively destroying the variable's predictive signal.

Table 2: Imputation comparison (quick RF, 5-fold stratified CV).

Method	F1 Macro	Std
Median/Mode	0.693	0.009
KNN (k=5)	0.686	0.011
MICE	**0.706**	0.013

Feature engineering. After MICE imputation and one-hot encoding of categorical variables (gender, platform, content type, activity type), the final feature set contains 30 variables. We added one interaction feature, `screen_time` × `late_night_usage`, motivated by the hypothesis that late-night social media use has a qualitatively different effect on mental health than daytime use. SHAP analysis later confirmed this interaction as the second most important predictor (§4.2).

3.4 Modeling

Six classifiers were trained for the primary classification task, including Logistic Regression, Decision Tree, Random Forest [5], Gradient Boosting, XGBoost, and a Stacking ensemble (RF + XGBoost base learners, Logistic Regression meta-learner), with all tree-based models using `class_weight='balanced'` to address the 7% High Risk minority class. We employed three validation strategies: (1) stratified 5-fold cross-validation, stratifying by class; (2) Leave-One-Dataset-Out (LODO), which trains on seven datasets and tests on the eighth, rotating through all eight to test genuine cross-dataset transfer; and (3) 80/20 stratified hold-out for final reported metrics. As a secondary track, Random Forest, Ridge, Lasso, and XGBoost regressors were trained on the continuous composite target, evaluated by RMSE, MAE, and R^2.

4. Results and Analysis

4.1 Classification Performance

Table 3: Classification results (5-fold stratified CV).

Model	Accuracy	F1 Macro	F1 Weighted
Logistic Regression	0.662	0.584	0.654
Decision Tree	0.711	0.628	0.704
Random Forest	0.808	**0.714**	0.805
Gradient Boosting	0.812	0.691	0.806
XGBoost	**0.818**	0.698	**0.813**
Stacking (RF+XGB)	0.676	0.600	0.670

Random Forest achieves the highest F1 macro (0.714), while XGBoost leads on accuracy (0.818). We selected RF as the primary model because F1 macro weighs all three classes equally, and with High Risk comprising only 7% of samples, a model that never predicts High Risk would still achieve 93% accuracy. The Stacking ensemble underperformed both of its base learners, likely due to insufficient base model diversity, as both RF and XGBoost are tree-based and produce correlated predictions, leaving the Logistic Regression meta-learner with little complementary information to exploit. On the 80/20 hold-out, RF achieved accuracy 0.806 and F1 macro 0.715, with High Risk as the most challenging class (precision 0.47, recall 0.49), reflecting the severe class imbalance.

4.2 Feature Importance and Interpretation

We used SHAP (SHapley Additive exPlanations) [3] to decompose each feature's contribution to individual predictions. Unlike Gini importance, SHAP provides both magnitude and direction of feature effects for each sample. The top predictors are `screen_x_latenight` (interaction), `late_night_usage`, `daily_screen_time_hours`, and `social_comparison`, in that order. The dominance of the interaction term over raw screen time indicates that the behavioral pattern matters more than the raw quantity, since scrolling at 2 AM carries a different risk profile than scrolling at 2 PM.

We also included `source_dataset` (ordinal encoded) as a feature in this analysis. It ranked 5th with a mean absolute SHAP value of 0.030, compared to 0.100 for the top feature. For comparison, an earlier approach using naive target construction on the same data found dataset identity as the 2nd most important feature, indicating that the model was largely predicting by dataset label rather than behavioral signal. Our theoretical range normalization substantially reduced this confounding, though it did not eliminate it entirely, and the residual signal primarily reflects the real-versus-synthetic distribution gap discussed in §4.3. The foundation model performed the SHAP computation and cross-fold consistency analysis; the human team interpreted the behavioral implications and connected the source_dataset ranking to the distribution shift phenomenon.

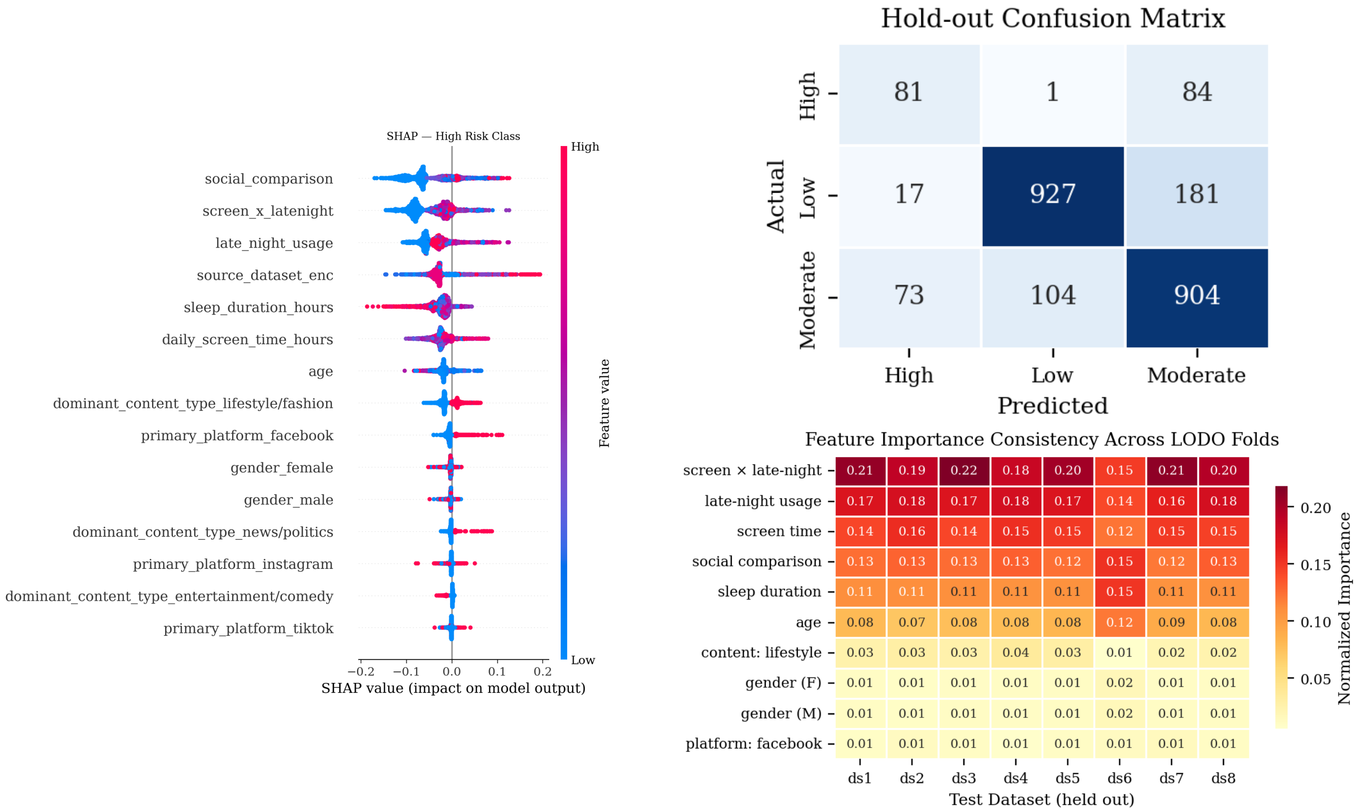

Figure 2: Left: SHAP beeswarm for the High Risk class, where each dot is one sample, color indicates feature value (red = high, blue = low), and horizontal position shows the SHAP contribution to the prediction. Upper right: Hold-out confusion matrix showing that High Risk is the most frequently misclassified class. Lower right: Normalized feature importance across all eight LODO folds, confirming that the top predictors remain stable regardless of which dataset is held out.

4.3 Cross-Dataset Generalization (LODO)

Table 4: LODO results. Train on 7 datasets, test on the held-out one.

Test DS	Type	N	Acc	F1m
ds1	Real	49	0.429	0.321
ds2	Synth	2,000	0.597	0.347
ds3	Synth	500	0.386	0.244
ds4	Real	481	0.320	0.292
ds5	Synth	20	0.150	0.130
ds6	Synth	8,000	0.626	0.440
ds7	Real	705	0.509	0.286
ds8	Synth	103	0.165	0.116
Mean			0.398	0.272

Standard 5-fold CV yields F1 macro of 0.714. LODO yields 0.272, a 62% drop. This gap reveals that within-sample performance substantially overstates generalization, and when the model faces an entirely new dataset with different survey design and population, performance collapses. Despite this overall drop, the feature importance structure remains remarkably stable across folds (Figure 2, right), with seven features appearing in the top-10 regardless of which dataset is held out.

4.4 Distribution Shift Diagnosis

To understand the LODO gap, we analyzed distributional differences between real and synthetic datasets. The target composite distributions diverge sharply. Real survey data has a mean of 0.553 with 32.1% High Risk, while synthetic data averages 0.326 with only 4.1% High Risk (KS statistic = 0.456, $p \approx 0$).

The root cause is self-selection bias [7]. Mental health surveys are voluntary; people who choose to participate tend to be those who already have mental health concerns, skewing real data toward higher severity. Synthetic data generators do not model this selection mechanism and instead sample from assumed population distributions, producing predominantly low-risk profiles. This is not a feature engineering problem or a model problem; it is a data composition problem that standard cross-validation, which shuffles all samples together, cannot reveal. Feature distributions are also shifted, with social comparison (KS = 0.594) and late-night usage (KS = 0.455) the most affected, and since these are also the top predictive features, the impact of distribution shift on model performance is amplified.

This diagnosis was conducted through a quantitative deep-dive with the foundation model performing KS tests, per-class breakdowns, and distribution overlays. The human team provided the domain interpretation, connecting the statistical evidence to the self-selection mechanism in voluntary surveys. This represents the most productive instance of FM-human collaboration in the project, where the FM identified *what* was happening quantitatively and the human explained *why*.

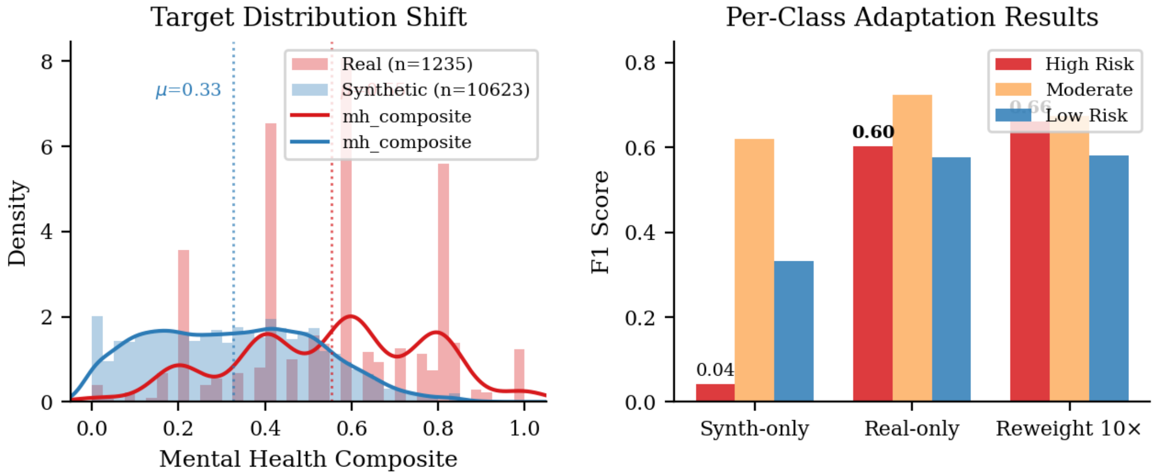

Figure 3: Left: Target distribution shift between real and synthetic data, with mean composites of 0.553 and 0.326 respectively, driven by self-selection bias in voluntary surveys. Right: Per-class F1 across adaptation strategies, showing that reweighting real data by 10× recovers High Risk detection from 0.04 to 0.66.

4.5 Adaptation Strategies

Knowing the root cause, we tested strategies to bridge the real-synthetic gap, evaluating on real data only (the deployment-relevant scenario).

Table 5: Adaptation results (evaluated on real data).

Strategy	F1m	High F1	Low F1	Mod F1
Synth-only	0.331	0.042	0.333	0.619
Real-only	0.634	0.603	0.577	0.723
All combined	0.619	—	—	—
Reweight (10×)	**0.639**	**0.661**	0.582	0.673
Warm-start	0.634	0.588	0.604	0.710

Reweighting real data by 10× achieves the best overall F1 macro (0.639) and dramatically improves High Risk detection from 0.04 to 0.66. The synth-only model essentially cannot detect high-risk individuals (F1 = 0.04), because it trained on a distribution where High Risk comprises only 4%. Reweighting outperforms real-only training for High Risk (0.661 vs. 0.603), suggesting that synthetic data provides complementary information when the model is guided to focus on the real signal through appropriate weighting. The tradeoff is a modest decrease in Moderate class F1 (0.723 → 0.673).

The warm-start experiment (pre-train on synthetic, fine-tune on real) yields feature importance rankings that correlate at $\rho = 0.736$ with the synthetic-only model, with the top three features identical. This confirms that synthetic data captures the correct feature relevance structure, namely *which* variables predict mental health risk, but cannot capture the target distribution shaped by self-selection, i.e., *how much* those variables matter at clinical severity levels.

Reweighting sensitivity analysis shows a gentle monotonic rise across weights from 1× to 50×, with the 8–20× range optimal. There is no sharp peak, indicating robustness to the specific weight choice.

4.6 Regression

As a secondary analysis, Random Forest regression on the continuous composite achieves $R^2 = 0.381$ and RMSE = 0.143. The moderate R^2 reflects the limited shared feature set (9 core features across 8 heterogeneous datasets) and the inherent noise in composite targets constructed from different instruments.

5. Discussion

5.1 Key Findings

Five findings emerge from this analysis.

1. **Late-night usage and social comparison** are the most robust predictors of mental health risk, stable across all eight LODO folds regardless of survey instrument or population.
2. **Behavioral pattern over raw quantity.** The screen-time-by-late-night interaction outranks raw screen time, suggesting that *when* and *how* people use social media matters more than *how much*.
3. **Cross-dataset generalization is the real challenge.** Standard CV (F1m = 0.714) dramatically overstates real-world performance compared to LODO (0.272).
4. **Self-selection bias** is the dominant distribution shift between real and synthetic data. This is a data composition problem, not a modeling problem.
5. **Synthetic data models feature structure, not target distribution.** Reweighting bridges the gap for minority-class detection.

5.2 Limitations

The High Risk class comprises only 7% of samples, limiting the model's ability to learn fine-grained risk patterns for the most clinically relevant group. Only 9 features are shared across most datasets; the remaining 21 are dataset-specific and contribute primarily through MICE imputation rather than direct signal. The composite target averages heterogeneous instruments with equal weight, potentially conflating different constructs of mental health. Datasets with single-variable composites (Datasets 1, 7, 8) have higher target variance. Finally, self-selection bias means that even our real data does not represent the general population; the model is trained on, and applicable to, survey-responding populations.

5.3 Implications

For digital wellness interventions, our results suggest targeting late-night usage and social comparison behaviors, as these are the most consistently predictive signals across diverse populations. For survey design, the sharp generalization gap underscores the need for standardized mental health instruments to enable cross-study comparison. For machine learning practice, LODO should be standard whenever multiple data sources are combined, as it is the only validation strategy that reveals distributional shifts masked by standard cross-validation.

Reflecting on our foundation model collaboration, the partnership was most effective when the FM handled systematic quantitative work (auditing 20+ correlations, computing SHAP across eight LODO folds) while the human team provided judgment and domain interpretation. In future work, we would provide structured, context-complete prompts from the outset rather than building context incrementally, and establish this division of labor as an explicit protocol earlier in the process.

The closing insight of this project concerns heterogeneity itself. We began by addressing measurement heterogeneity, asking how to unify eight different instruments onto a common scale. We ended by discovering that the more consequential heterogeneity is in sampling, because who participates in each survey shapes the target distribution more than any modeling choice. Measurement heterogeneity is an engineering problem with a principled solution. Sampling heterogeneity is a scientific problem that requires understanding the data generation process.

6. Conclusion

We presented a pipeline for multi-dataset mental health risk prediction that addresses the challenge of heterogeneous instruments through theoretical range normalization. While the resulting model achieves reasonable within-sample performance (F1 macro = 0.714), LODO evaluation reveals a sharp generalization gap traced to self-selection bias in voluntary surveys. Reweighting real data recovers minority-class detection, and synthetic data is shown to capture feature relevance structure while missing the target distribution shaped by survey participation patterns. The harder heterogeneity in multi-source mental health research is not in how you measure, but in who you measure.

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